



- c) Magnel-Balton system  
d) Long line method
- Q.6 During erection, the pile of length " $l$ " is supported by a crane at a distance of \_\_\_\_\_ from the driving end of pile which rests on the ground.  
a)  $0.207l$                       b)  $0.707l$   
c)  $0.293l$                       d)  $0.793l$
- Q.7 The best suitable method which takes care about safety and serviceability is \_\_\_\_\_.  
a) Working Stress Method  
b) Limit State Method  
c) Ultimate Load Method  
d) None of these
- Q.8 In limit state design, permissible bond stress in case of deformed bars is more than that in plain bars by \_\_\_\_\_.  
a) 25%                      b) 40%  
c) 50%                      d) 60%
- Q.9 Maximum diameter of steel bar available as per IS 1786:2008 is \_\_\_\_\_.  
a) 40mm                      b) 45mm  
c) 50mm                      d) None of these
- Q.10 Stress at Neutral Axis is \_\_\_\_\_.  
a) Compressive              b) Tensile  
c) Shearing                  d) Zero

### SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Pile foundation is an example of \_\_\_\_\_ foundation. (Deep / Shallow)

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- Q.12 In post-tensioned members reinforcement is tensioned \_\_\_\_\_ the concrete has fully hardened. (Before / After)
- Q.13 Minimum grade of concrete required for pre-tensioning is \_\_\_\_\_. (M-25 / M-40)
- Q.14 Pre-stressing helps in avoiding the formation of \_\_\_\_\_ in tensile zone. (Cracks / Bonding)
- Q.15 More the diameter of tendon \_\_\_\_\_ is the ultimate strength (Higher / Lesser)
- Q.16 The downward drag acting on a pile due to movement of surrounding is called \_\_\_\_\_ (Sinking of pile / Negative skin friction)
- Q.17 A raft foundation is also known as \_\_\_\_\_ (Mat foundation / pile foundation)
- Q.18 The pre-stressing force \_\_\_\_\_ with the time. (Increase / Decrease)
- Q.19 The piles having one or more bulbs are called \_\_\_\_\_ (under reamed piles / pile cap)
- Q.20 A vertical column of relatively large diameter than a pile is termed as a \_\_\_\_\_ (pile cap / pier)

### SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Enlist the techniques of pre tensioning.
- Q.22 Write a short note on small diameter bored cast-in-situ piles.
- Q.23 Differentiate between the pre-tensioning method and post-tensioning method of concrete. (Any five)
- Q.24 Write the specifications of admixtures as per IS code, used in pre stressed concrete.

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